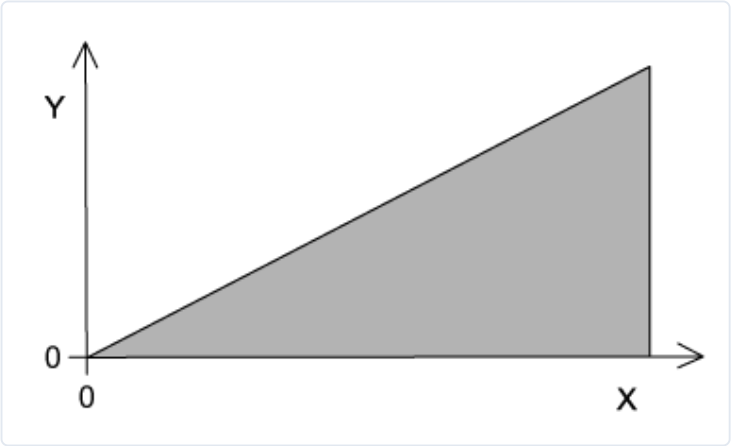
CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

6.2 Elastic & Plastic Behaviour

Syllabus Section: 6. Deformation of Solids **Paper Type:** Multiple Choice (Paper 1) **Total:** 25 Questions (25 Marks)

Question 1 Easy [1 mark]

A student experimented on a metal wire; the graph below was plotted.



The area under the graph shows the total strain energy stored in the wire when stretched.

How should the axes be labelled?

	Y	X
A	mass	extension
B	force	extension
C	strain	energy
D	stress	strain

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Strain is the measurement of how much an object is stretched
- Strain energy is equal to $\frac{1}{2}k \quad x^2 = = \frac{1}{2}Fx$
- The question states that the area under the graph shows the total strain energy stored
- For this to be true, the axis must be:
    - X = extension *x*

- Y = force *F*

Question 2 Easy [1 mark]

Four rods were made from four different materials; all the rods had the same dimensions.

At room temperature, which of the following would sustain the biggest plastic deformation?

A the ductile material steel

B the brittle material glass

C the brittle material carbon

D the ductile material aluminium

Mark Scheme & Explanation:

*The correct answer is **D** because:*

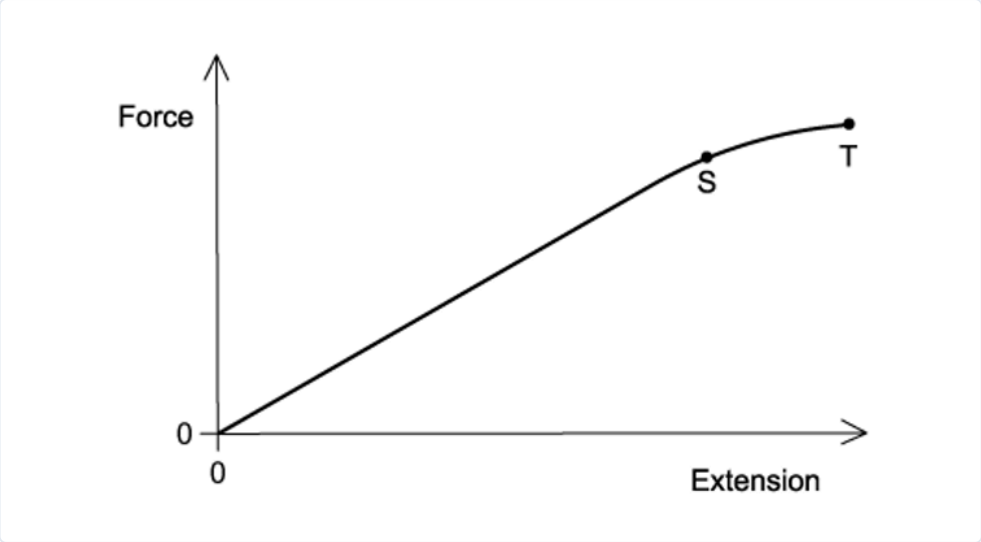
- Brittle materials break easily under tension
- Ductile materials are solids that can deform under tensile stress
- Aluminium can sustain a larger plastic deformation than steel, so the correct answer is **D**

A is incorrect because steel is ductile, but aluminium can sustain a larger plastic deformation

B & **C** are incorrect because glass and carbon are examples of brittle materials and will break easily with a small tension

Question 3 Easy [1 mark]

A student drew a force-extension graph for a sample of metal. She subjected the sample to a force that increased to a maximum then decreased to zero.



When the sample contracts it follows the same force-extension curve as when it was being stretched.

What is the behaviour of the metal between S and T?

A both elastic and plastic

B elastic but not plastic

C not elastic and not plastic

D plastic but not elastic

Mark Scheme & Explanation:

*The correct answer is **B** because:*

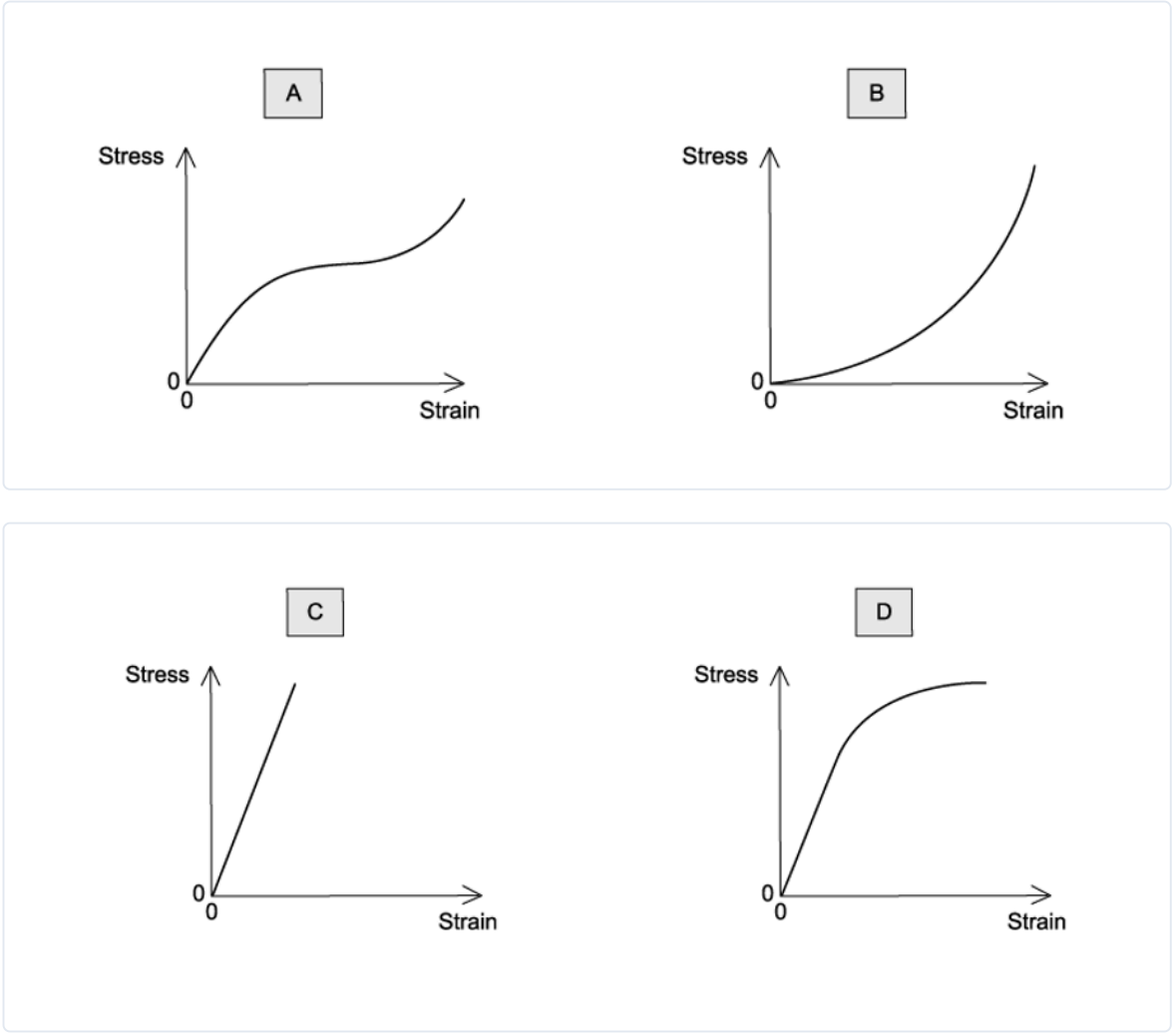
- Force-extension graphs tell us about the strain on a material
- Elasticity is the ability of the material to resist a permanent change, that is it will return to its original shape and size after the stress is removed
- Plasticity is a non-reversible change of shape in response to a force
    - Point S on the graph is the limit of proportionality
    - Point T is the elastic limit

- After the elastic limit, the material would exhibit plastic behaviour, meaning it would have a permanently changed shape
- Therefore, as the sample has not gone past point T, the material would be exhibiting elastic behaviour, but not plastic behaviour

Question 4 Easy [1 mark]

Graphs of the stress-strain for four different materials are shown below.

Which one shows the correct graph for a ductile metal ?



A

B

C

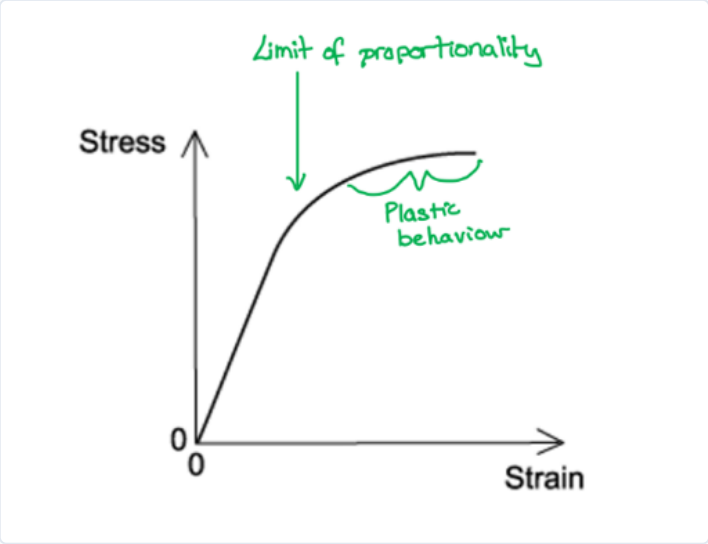
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Brittle materials break easily under tension
- Ductile materials are solids that can deform under tensile stress
- A ductile material will withstand large plastic strain, whereas a brittle material will not

- A metal wire tends to be described as ductile as it will experience a large amount of extension when subjected to stress, especially beyond the elastic limit



- A ductile metal will obey Hooke's law initially, which is represented as a straight line on the graph
- The metal would then exhibit plastic behaviour after the limit of proportionality, which is represented on the graph as a curve

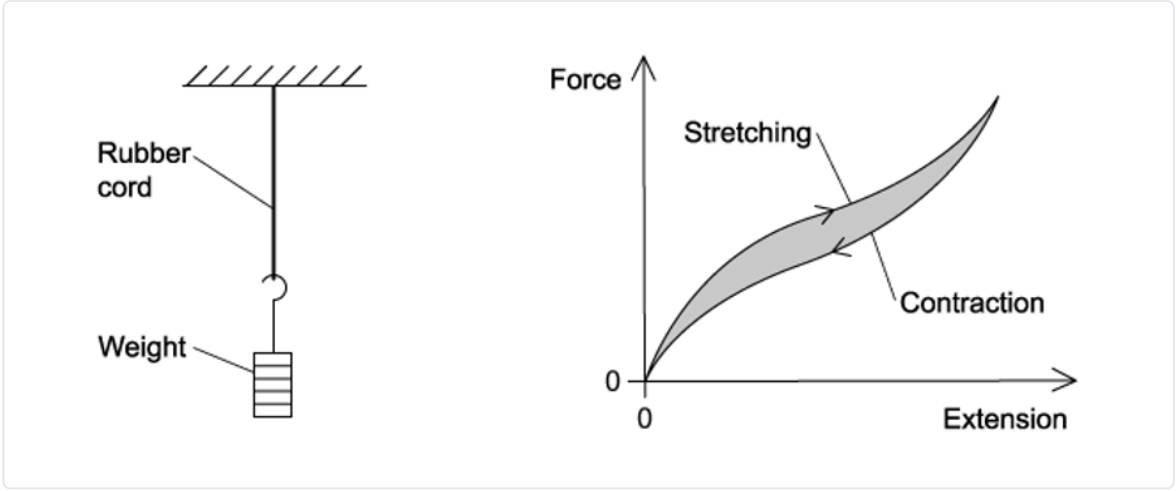
A is incorrect because the graph shows an unusual shape stress-strain graph associated with a polymer, such as rubber

B is incorrect because the graph doesn't show proportionality

C is incorrect because the graph shows a brittle material that shows no plastic strain

Question 5 Easy [1 mark]

A weight was attached to the lower end of a length of rubber. The rubber was hung from a solid support. The weight was gradually increased from zero then gradually reduced to zero.



The graph shows the force-extension curve for contraction (below) and the force-extension curve for stretching.

What does the shaded area between the curves represent?

A the amount of elastic energy stored in the rubber

B the work done by the rubber cord during contraction

C the work done on the rubber cord during stretching

D the amount of thermal energy dissipated in the rubber

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The area under the force-extension graph shows the strain energy (work done)
  - Strain energy is equal to $= \frac{1}{2}Fx$
- The area under the stretching graph is the energy used in stretching the rubber
- The area under the contraction graph is less than the area of the stretching graph
- The difference is shown in the shaded area between the two
  - This is the thermal energy which is dissipated to the surroundings

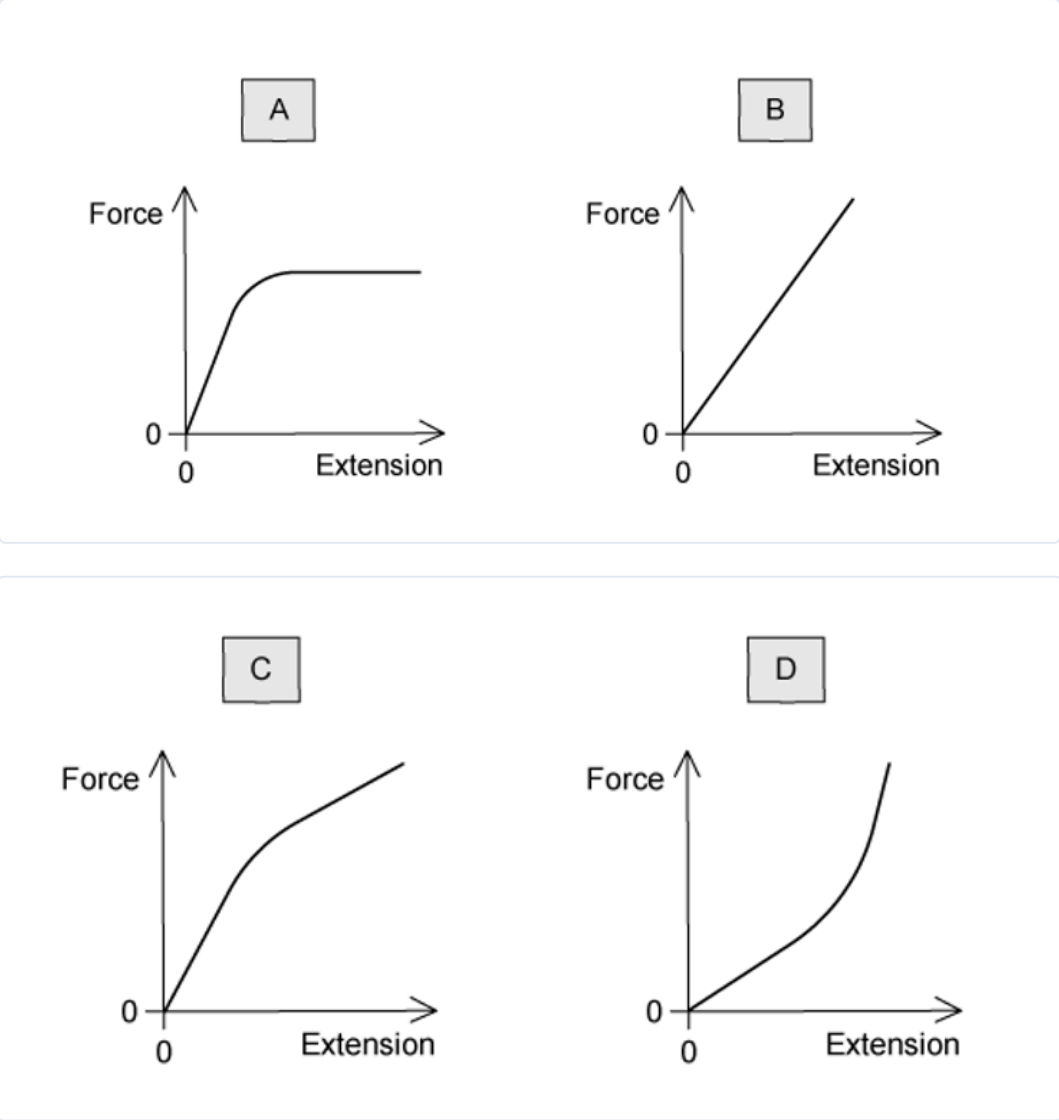
A is incorrect because the amount of elastic energy stored in the rubber is the total area under both curves

B is incorrect because the work done by the rubber cord during contraction is represented by the area under the lower curve

C is incorrect because the work done on the rubber cord during stretching is represented by the area under the upper curve

Question 6 Easy [1 mark]

The graphs below are force-extension graphs, which one is a rubber band this is stretched almost to its breaking point ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The force-extension graph shows the strain energy (work done)
  - Strain energy is equal to $\frac{1}{2}Fx$

- Rubber is a polymer and when fully stretched the long-chain molecules are fully lined up so the rubber will not stretch any more
    - Rubber deforms elastically up to a certain point as the polymers chains, which were previously curled up, uncurl
- Graph **D** shows that once the band is stretched fully, it takes lots of force to increase the extension

A is incorrect because the graph suggests that once stretched no extra force is needed to extend the rubber

B is incorrect because the graph shows a material that obeys Hooke's law

C is incorrect because the graph suggests that less force is needed to extend the rubber band even for a small extension

Question 7 Easy [1 mark]

A metal wire had a tension *F* and extension *x* with a Young modulus of *E*; the wire obeyed Hooke's law

Which expression represents the elastic strain energy stored in the wire?

A $\frac{1}{2}Fx$

B Ex

C $\frac{1}{2}Ex$

D Fx

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The question states that Hooke's law is obeyed so that graph would be a straight line passing through the origin
- The triangular area under the graph represents the elastic strain energy
- Therefore, the equation for elastic strain energy is $= \frac{1}{2}Fx$

Question 8 Easy [1 mark]

Which of the following is the correct definition for elastic and plastic behaviour?

	Elastic behaviour of a material	Plastic behaviour of a material
A	returns to its original shape and size	suffers permanent deformation
B	obeys Hooke's Law	extends continuously under a steady load
C	has a linear force-extension curve	has a horizontal force-extension curve
D	extends only within the limit of proportionality	extends beyond the limit of proportionality

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Elasticity is the ability of the material to resist a permanent change, that is it will return to its original shape and size after the stress is removed
    - Hence, an elastic material will return to its original shape and size
- Plasticity is a non-reversible change of shape in response to a force
    - Hence, a plastic material will suffer a permanent deformation

B is incorrect because materials with plastic behaviour will not extend continuously

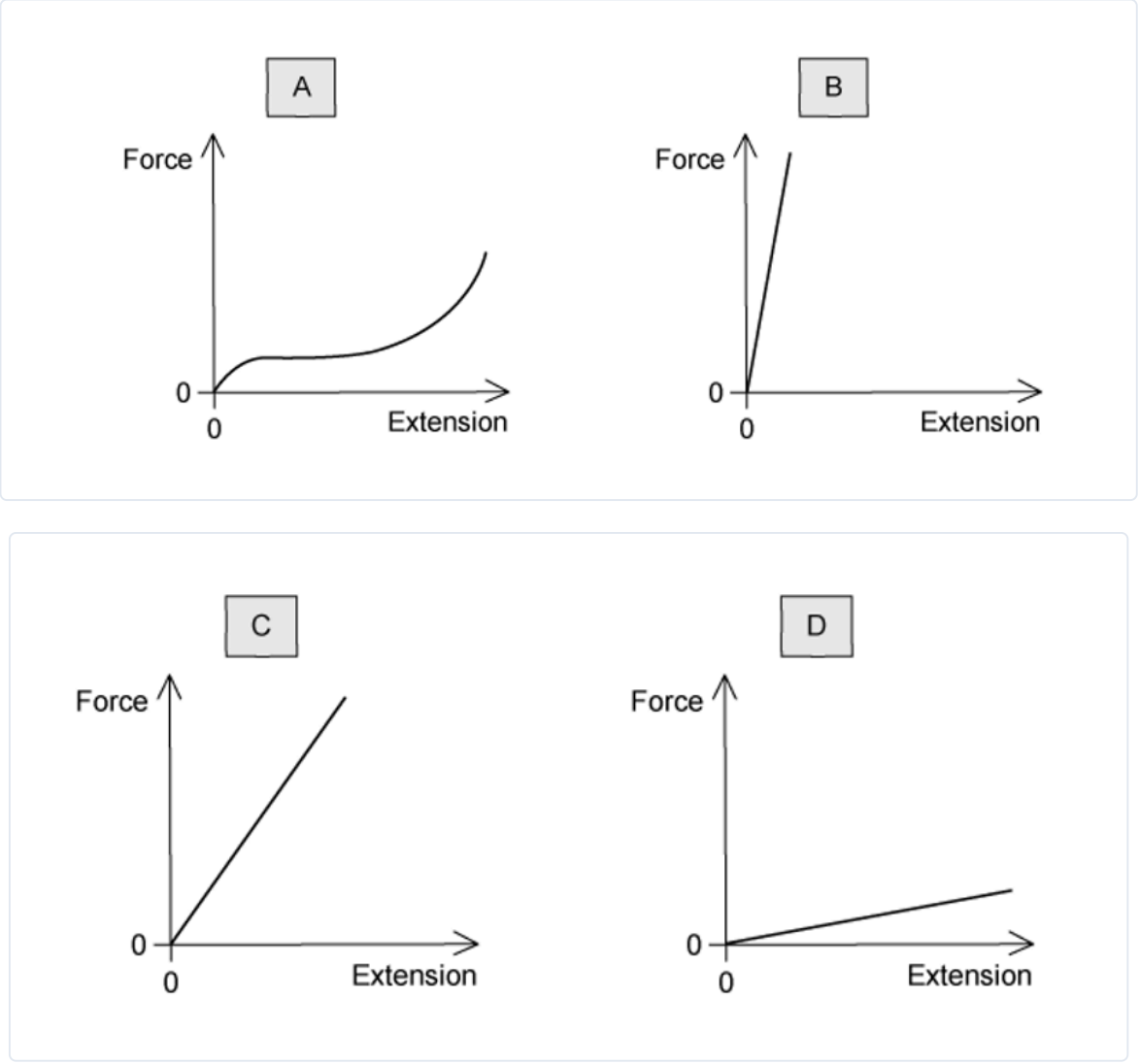
C is incorrect because not all elastic materials will show a linear force-extension curve, consider rubber

D is incorrect because the definition of a material obeying Hooke's law is that extends only within the limit of proportionality

Question 9 Easy [1 mark]

The graphs below all show force-extension graphs; they are all drawn to the same scale.

Which graph shows the object that has been deformed and has the greatest amount of elastic potential energy ?



A

B

C

D

Mark Scheme & Explanation:

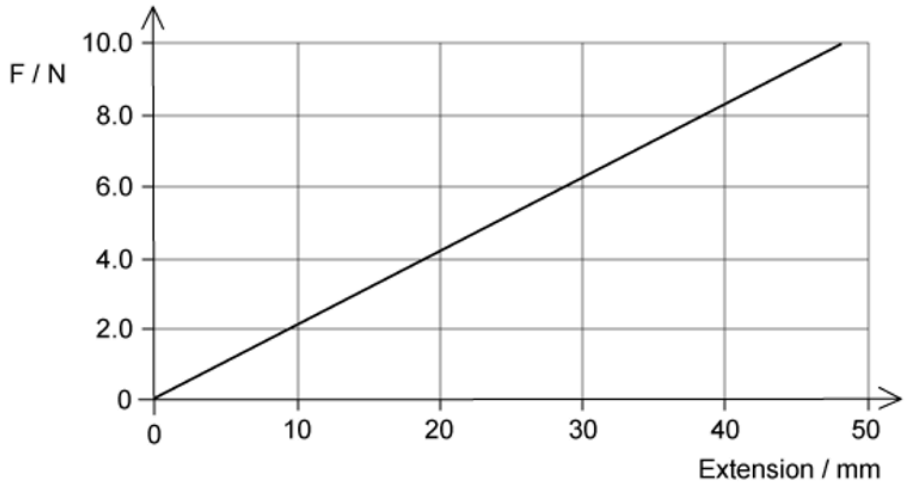
The correct answer is ***C*** *because:*

- Elastic potential energy is equal to $\frac{1}{2}Fx$
- It is represented by the area under the force-extension graph

- The graph that shows the largest area under the line or curve is **C**

Question 10 Easy [1 mark]

A spring is extended with varying forces; the graph below shows the results.



What is the energy stored in the spring when the extension is 30 mm?

A 0.25 J

B 0.95 J

C 0.095 J

D 0.19 J

Mark Scheme & Explanation:

*The correct answer is **C** because:*

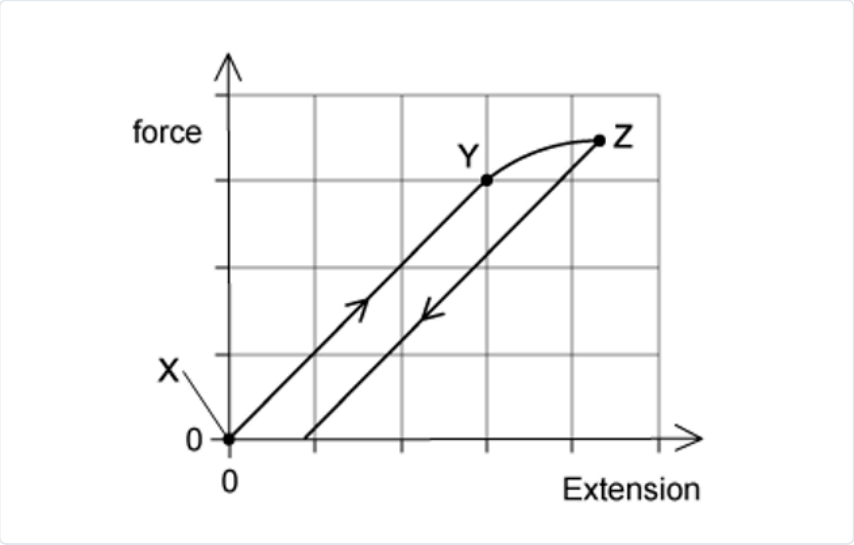
- Elastic potential energy $\frac{1}{2}Fx$
- From the graph
  - When the extension is x = 0.03 m
  - Force is equal to F = 6.3 N
  - Strain energy $= \frac{1}{2} \times 6.3 \times 0.03 = 0.0945$ J
- Therefore, the energy stored in the spring is 0.095 J

Question 11 Medium [1 mark]

A student was investigating the extension in a long thin metal wire. They suspended the wire from a fixed support, and hung it vertically then hung masses from the lower end.

The load was increased and decreased from zero to the maximum and back again.

The graph shows the force-extension of the wire.



Where on the graph would the elastic limit be found?

A exactly at point Z

B exactly at point Y

C just beyond point Y

D anywhere between point X and point Y

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The elastic limit is just beyond the limit of proportionality
  - If you increase the load beyond this point, the material will deform plastically
- The limit of proportionality is shown by point Y
  - This is the point after which Hooke's Law is no longer obeyed
- Therefore, the elastic limit is just beyond point Y

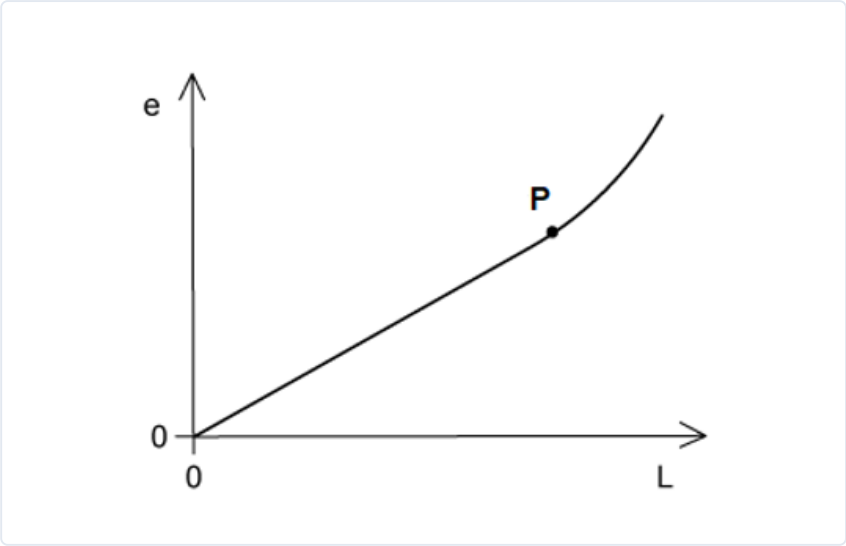
A is incorrect because point Z is beyond the elastic limit since plastic deformation occurs when the force is removed

B is incorrect because Y shows the limit of proportionality

D is incorrect because between point X and Y the force is proportional to the extension

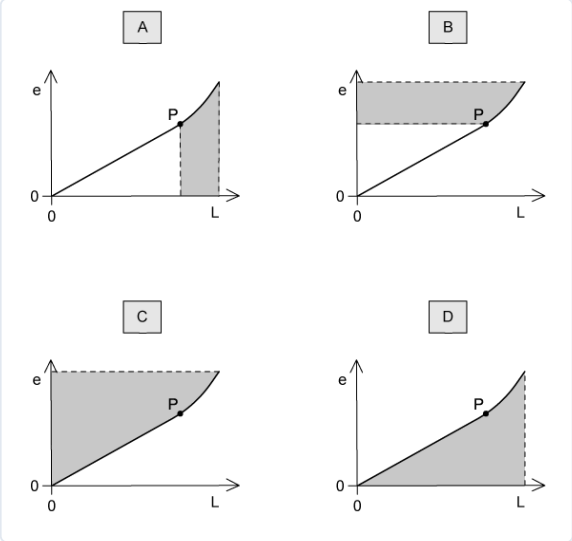
Question 12 Medium [1 mark]

A rod is extended with a force so that its length increases. The variation with load *L* of the extension *e* of the rod is shown in the graph.



Point **P** is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The elastic limit is shown by point P
    - This is the point at which if you increase the load any further, the material will deform plastically
- Therefore the plastic deformation is beyond point P
- The work done will be the area under the line in a load-extension graph
    - In this case, the extension *e* is on the y-axis and the load *L* is on the x-axis, so we require the area between the curve and the y-axis

A is incorrect because plastic deformation is the area under a force-extension graph not extension-force

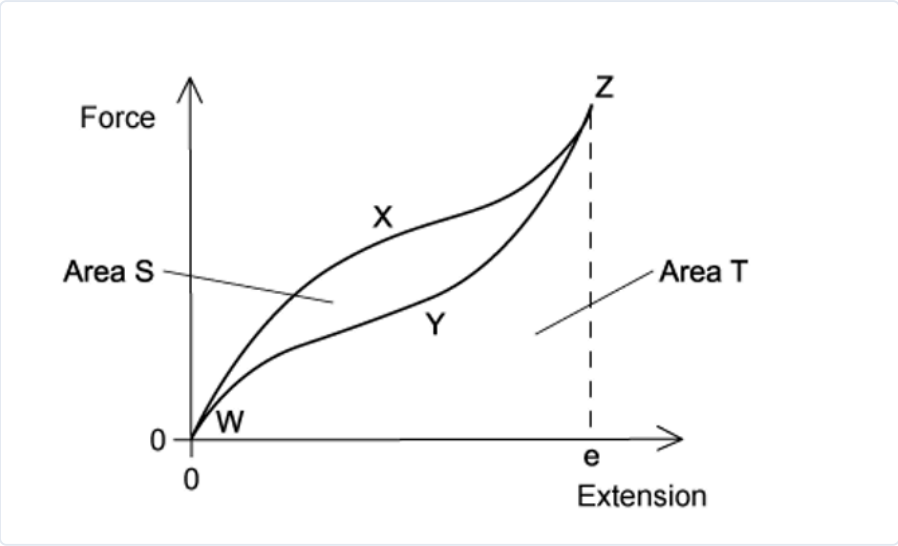
C is incorrect because the graph shows the work done for the whole extension not just plastic deformation

D is incorrect because work done is the area under a force-extension graph not extension-force

Exam Tip: always check the axes of the graph, many exam questions will change the axes to a form you are not used to in order to catch you out, so look out!

Question 13 Medium [1 mark]

A student was investigating a rubber band when stretched and relaxed to its original length. The graph shows the force-extension of the rubber band.



As the force is increased, the curve follows the path WXZ to extension *e*. As the force is reduced, the curve follows the path ZYW to return to zero extension.

The area labelled S is between the curves WXZ and ZYW. The area labelled T is bounded by the curve ZYW and the horizontal axis.

Which statement about the process is correct?

A area S is the energy which heats the band as it is stretched to *e*

B area S is the elastic potential energy stored in the band when it is stretched to *e*

C (area S + area T) is the minimum energy required to stretch the band to *e*

D area S is the spring constant of the band at e

Mark Scheme & Explanation:

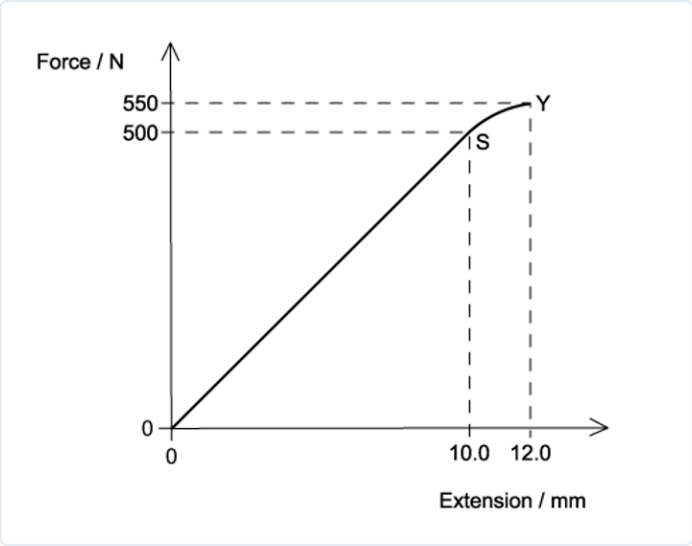
*The correct answer is **C** because:*

- The area under the curve OXZ is the work done in stretching the rubber band
  - Area S represents the amount of thermal energy dissipated to the surroundings
  - Area T represents the energy that is stored at maximum extension
- Therefore, the minimum energy required to stretch the band to *e* is equal to = (area S + area T)

A, B & D are incorrect because the area S represents the amount of thermal energy dissipated to the surroundings by the rubber band

Question 14 Medium [1 mark]

A sample of metal was stretched until it started to go into plastic deformation. The graph below shows the behaviour of the metal.



When the sample was stretched from 0 to 12.0 mm extension, what is the work done? Assume that the curve SY is a straight line to simplify the calculation

A 6.60 J

B 3.60 J

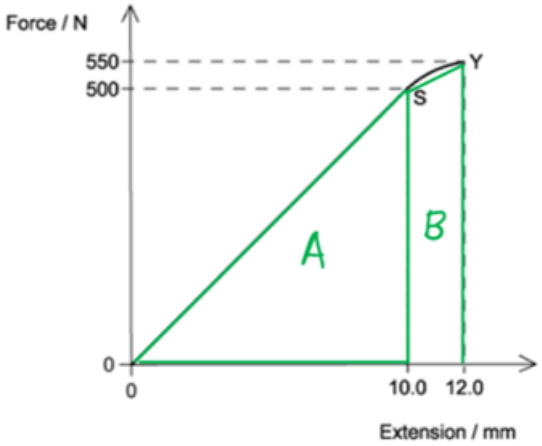
C 3.55 J

D 3.30 J

Mark Scheme & Explanation:

*The correct answer is **C** because:*

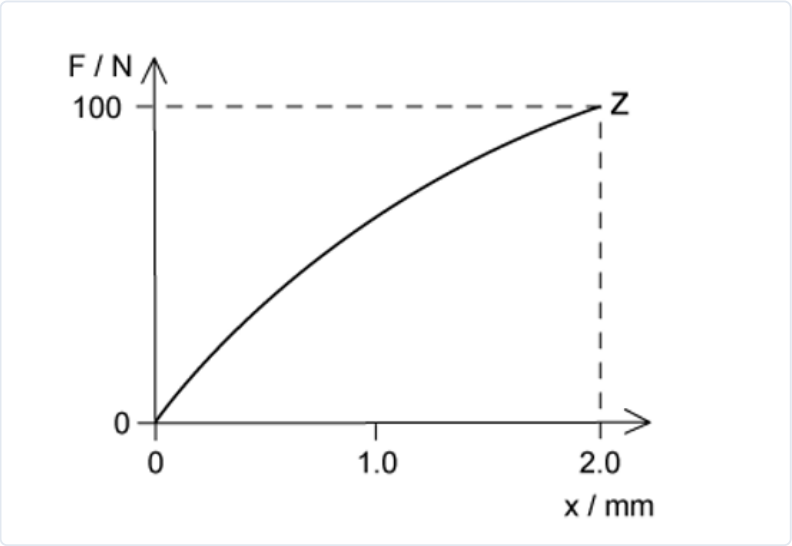
- Work done = force × distance = area under the graph
- Divide the total shape into a triangle and a trapezium



- Area of shape A = $\frac{1}{2} \times 500 \times 0.010 = 2.50$ J
- Area of shape B = $\frac{1}{2} \times 500 + 550 \times 0.002 = 1.05$ J
    - Therefore, total area $= 2.50 + 1.05 = 3.55$ J

Question 15 Medium [1 mark]

A wire made from a new composite material was loaded with masses and the force-extension curve drawn.



What could be the value of the strain energy stored in the wire when it is stretched elastically to point Z ?

A 0.20 J

B 0.11 J

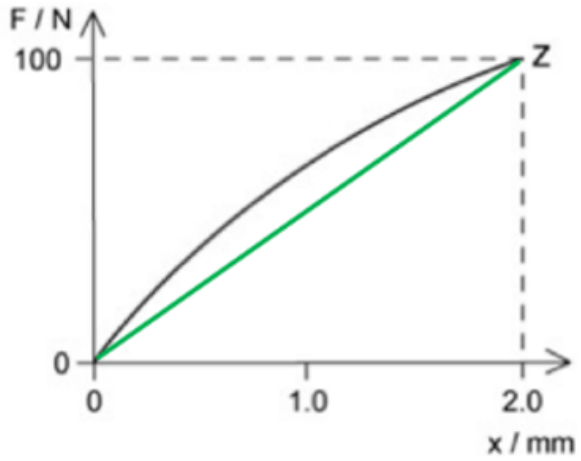
C 0.10 J

D 0.09 J

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The area under a force-extension curve gives strain energy
- Draw a straight line drawn from (0, 0) to point Z



- Calculate the area of the triangle
  - Strain energy $= \frac{1}{2}Fx = \frac{1}{2} \times 100 \times 2 \times 10^{-3} = 0.1$ J
- This is less than the area under curve shown in the graph
- The answer should be about 10% greater to account for the curve = 0.11 J

A is incorrect because this is twice the calculated value so too big

C & **D** are incorrect because the actual area under the curve should be greater than the figure calculated

Question 16 Medium [1 mark]

A student was investigating a spring with a spring constant *k* of 3.0 N cm^{-1}, the spring was stretched over a range in which elastic deformation occurred.

Which row, for the stated applied force, gives the correct extension and strain energy?

	Force / N	Extension / cm	Strain energy / mJ
A	24.0	8.0	960
B	12.0	3.0	200
C	6.0	2.0	120
D	3.0	1.0	1.5

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Using Hooke's law $F = kx$
  - Where *k* is the spring constant, 3.0 N cm^{-1}, or 300 N m^{-1}
- Strain energy = $\frac{1}{2}Fx$
- Consider option **A**:
  - Force = 24 N, extension = 0.08 m
  - Strain energy $= \frac{1}{2} \times 24 \times 0.08 = 0.96J = 960$ mJ

B is incorrect because force = 12 N, extension = 0.03 m

strain energy = 0.5 × 12 × 0.03 = 0.18 J = 180 mJ

C is incorrect because force = 6 N, extension = 0.02 m

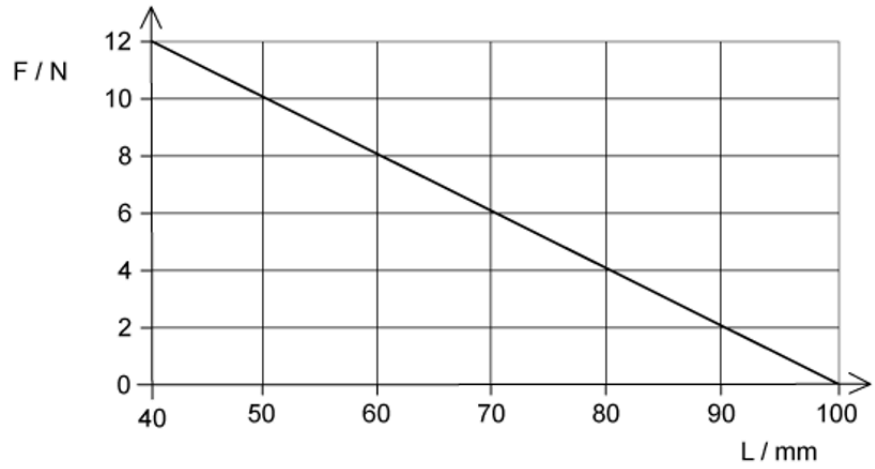
strain energy = 0.5 × 6 × 0.02 = 0.06 J = 60 mJ

D is incorrect because force = 3.0 N, extension = 0.01 m

strain energy = 0.5 × 3 × 0.01 = 0.015 J = 15 mJ

Question 17 Medium [1 mark]

A force compresses a mattress spring by 100 mm. The graph shows the effect of the force *F* on the length *l* of the spring.



What is the energy stored in the spring when the length is 70 mm?

A 0.63 J

B 0.27 J

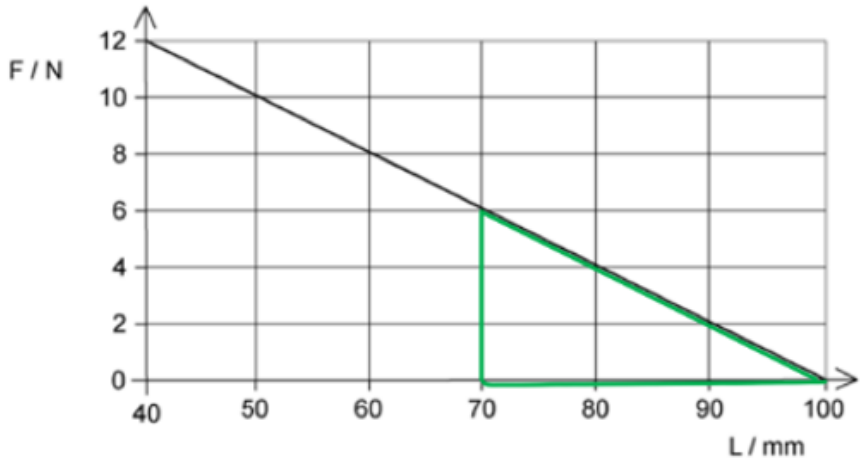
C 0.21 J

D 0.090 J

Mark Scheme & Explanation:

*The correct answer is **D** because:*

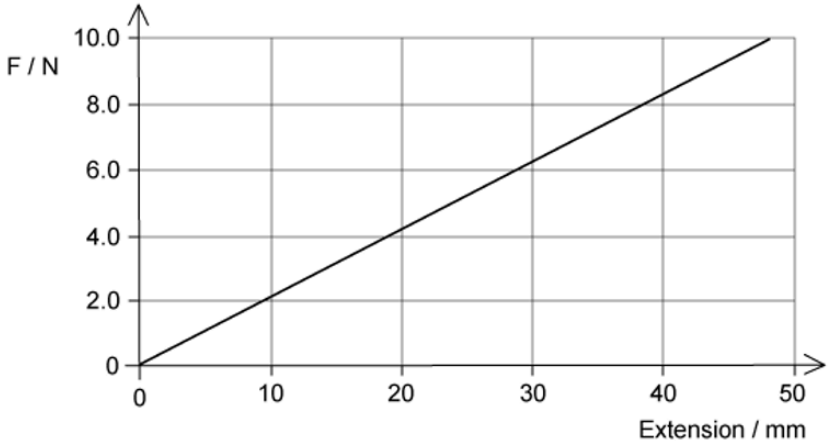
- The original length of the spring is 100 mm this is shown on the graph where the force is zero
- The energy stored when the spring is compressed to 70 mm is equal to the area under the graph
    - Therefore, compression = 100 – 70 = 30 mm = 0.03 m



- Elastic potential energy = $\frac{1}{2}Fx$
  - From the graph:
  - When compression = 100 – 70 = 30 mm = 0.03 m
  - Force = 6 – 0 = 6.0 N
- Therefore, strain energy $= \frac{1}{2} \times 6.0 \times 0.03 = 0.090$ J

Question 18 Medium [1 mark]

A spring is extended due to a force. The graph shows the effect of changing force on the extension of the spring.



What is the elastic potential energy of the spring with a force of 8.0 N ?

A 152 mJ

B 58 J

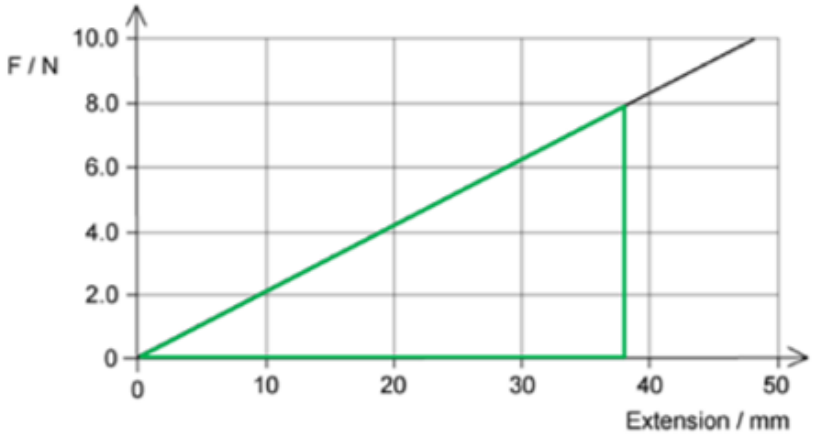
C 85 mJ

D 5.8 mJ

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The elastic potential energy is the area under the force-extension graph.
- The question asks for the elastic potential energy when a force of 8.0 N is applied

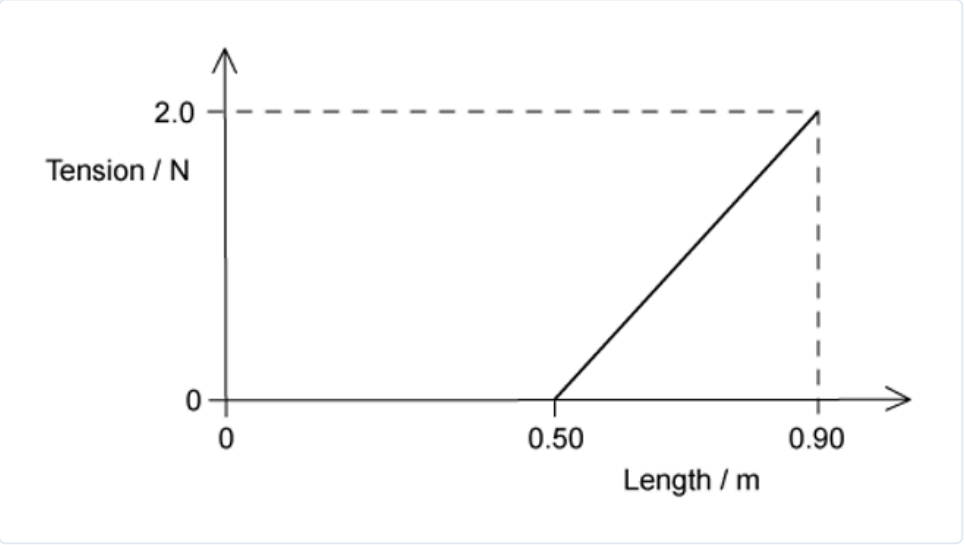


- Elastic potential energy $= \frac{1}{2}Fx$
- Elastic potential energy $= \frac{1}{2} \times 8.0 \times 0.038 = 0.152$ J *or* 152 mJ

Question 19 Medium [1 mark]

A force of 2.0 N is applied to a spring of unextended length 0.50 m and has a new length of 0.90 m.

A graph shows the variation of its length with tension.



How much strain energy is stored in the spring ?

A 1.8 J

B 0.90 J

C 0.40 J

D 0.80 J

Mark Scheme & Explanation:

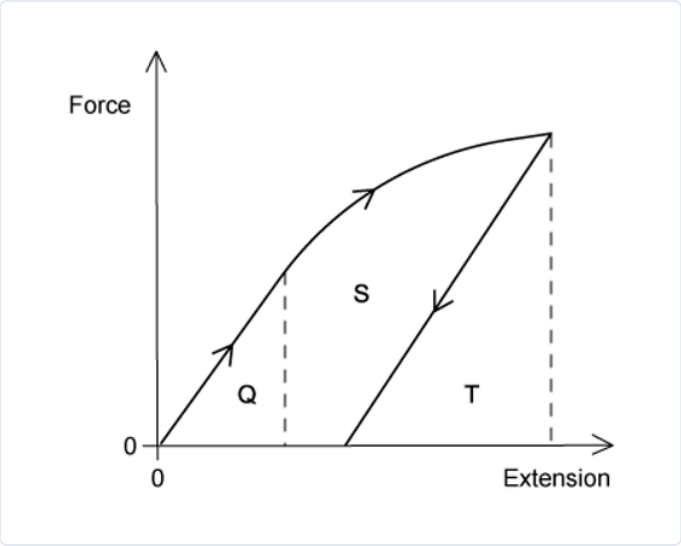
*The correct answer is **C** because:*

- Energy stored in a spring is equal to $= \frac{1}{2}Fx$
- The area under the tension-length graph gives the energy stored in the spring
  - Therefore, strain energy $= \frac{1}{2} \times 2.0 - 0 \times 0.9 - 0.5 = 0.40$ J

Exam Tip: Strain is equal to $\frac{extension}{original\ \ length}$ this is not the same as the energy stored in the spring

Question 20 Medium [1 mark]

A student carried out an experiment on a ductile material; stretching it with a tensile force beyond its elastic limit. The force is removed.



The graph shows the force-extension of this experiment.

Which area represents the net work done on the sample?

A *T*

B *S* + *T*

C *Q* + *S*

D *Q*

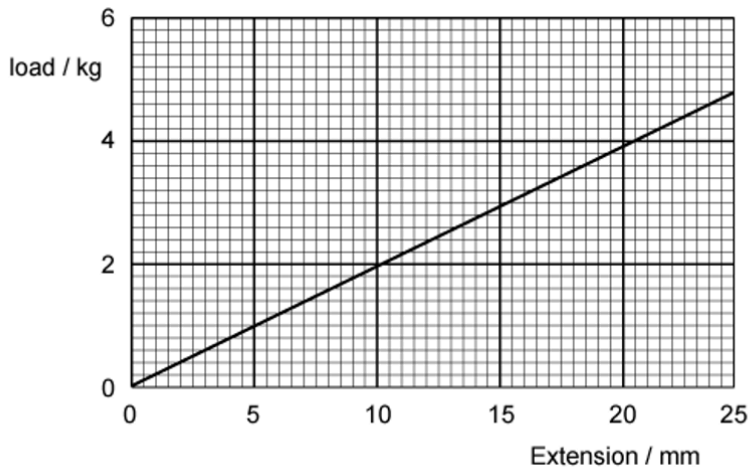
Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Strain energy $= \frac{1}{2}Fx$
    - So, the area under the curve in a force-extension graph equals the work done
- *Q*, *S* and *T* show sections of the area under the line
- The materials behaviour is shown as the force is applied and then removed
    - Therefore, to calculate the work done the total area enclosed by the curve is needed so *Q* + *S*

Question 21 Hard [1 mark]

A wire undergoes elastic deformation; the graph below shows the load-extension graph.



How much work is done on the wire to increase the extension from 10 mm to 20 mm?

A 0.18 J

B 0.28 J

C 0.37 J

D 0.59 J

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Elastic strain energy $= \frac{1}{2}Fx$
  - Therefore, the area under the force-extension graph equals the work done
- The applied load needs to be converted from kg to N by multiplying by g = 9.81 m s^{-2}
- The work done is equal to the difference in the elastic strain energy between extensions 10 mm and 20 mm
  - At extension 10 mm (0.01 m), load = 2.0 × g
  - At extension 20 mm (0.02 m), load = 3.9 × g
- Work done = $\frac{1}{2}$× [(3.9 × 9.81 × 0.02) – (2.0 × 9.81 × 0.01)] = 0.284 J

- Therefore, the work is done on the wire is 0.28 J, which is option **B**

Question 22 Hard [1 mark]

A material with elastic properties that obeys Hooke's law has a Young modulus *E* and has a tensile stress of *S* applied.

What is the expression for the elastic energy stored per unit volume of the material?

A $\frac{E}{2s^2}$

B $\frac{S^2}{E}$

C $\frac{S^2}{2E}$

D $\frac{2E}{S^2}$

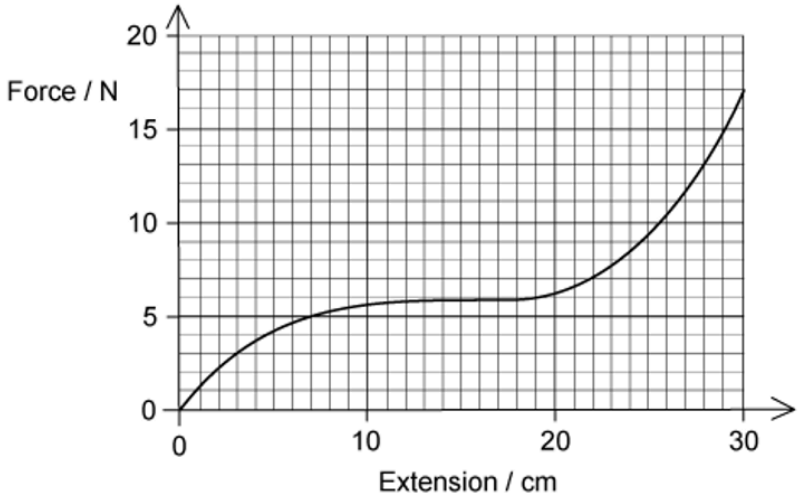
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Elastic energy $= \frac{1}{2}Fx$
- Elastic energy per unit volume $= \frac{Fx}{2V}$
- Volume $= AL$, so, substituting this into the elastic energy per unit volume equation:
  - Elastic energy per unit volume $= \frac{Fx}{2AL} = \frac{x}{2L}\frac{F}{A}$
  - Tensile stress *S* is equal to $= \frac{F}{A}$
  - So, the elastic energy per unit volume $= \frac{xS}{2L}$
- Strain is equal to $= \frac{x}{L}$
  - So, elastic energy per unit volume $= \frac{S}{2}\frac{x}{L}$
- Young modulus $E = \frac{stress}{strain}$
  - Therefore, strain $= \frac{x}{L} = \frac{S}{E}$
- Combining the volume and strain equations
  - Elastic energy per unit volume $= \frac{s}{2}\frac{S}{E} = \frac{S^2}{2E}$

Question 23 Hard [1 mark]

A graph was drawn of the effect on the length of applying a force to a rubber cord.



What is the maximum strain energy in this deformed rubber cord?

A 200 J

B 5.1 J

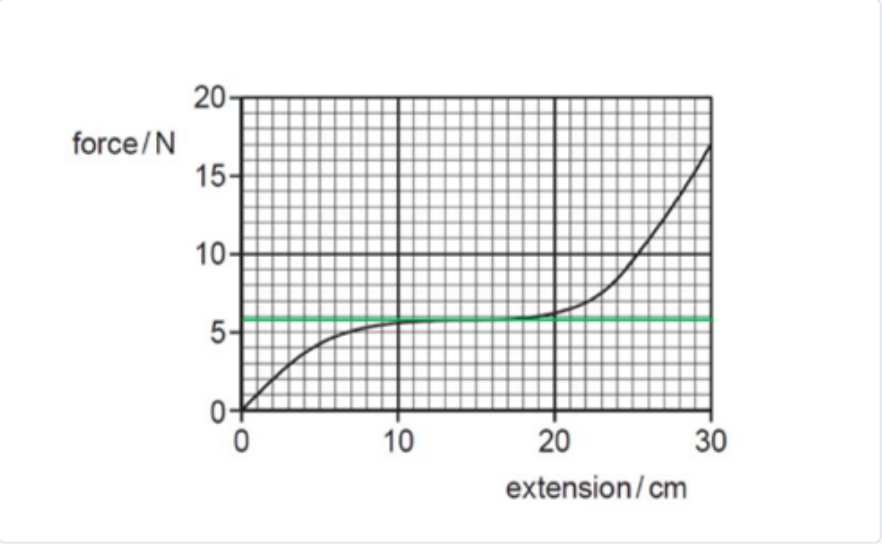
C 2.5 J

D 1.9 J

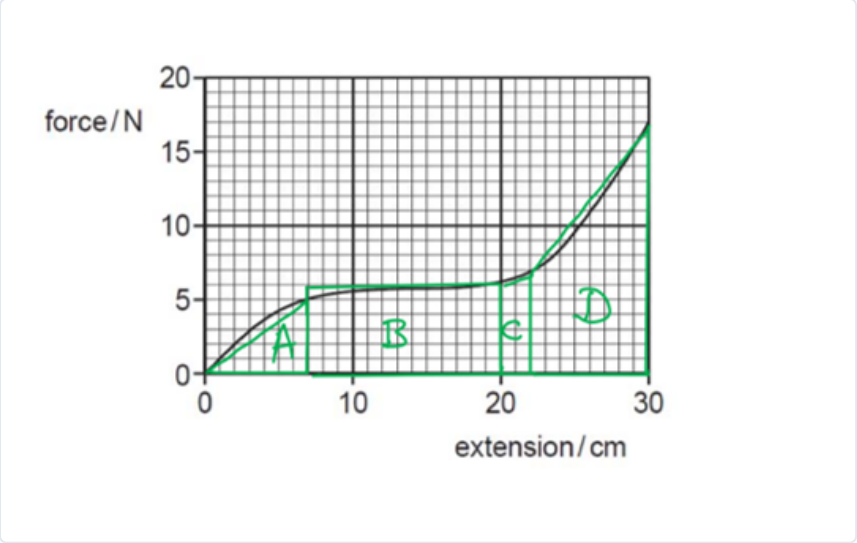
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The area under the force-extension graph gives the energy stored in the rubber cord
- Method 1: Draw a straight line on the graph to give the approximate mean force
    - Giving 6 N × 0.3 m = 1.8 J



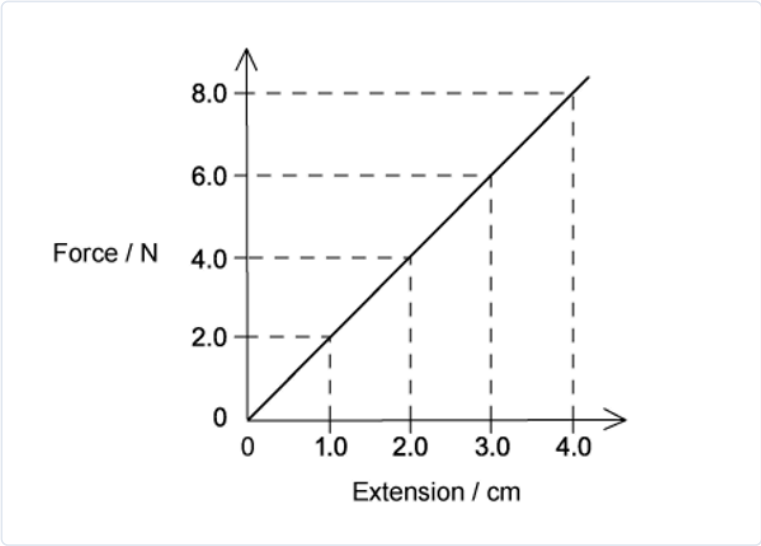
- Method 2: Split the area into a series of triangles and rectangles to work out the total area



  - Area A = ½ × 0.07 × 5 = 0.175J
  - Area B = (0.20 – 0.07) × 5.5 = 0.715 J
  - Area C = ½ × (5.5 + 6) × 0.02 = 0.115 J
  - Area D = ½ × (6 + 17) × (0.30 - 0.22) = 0.92 J
  - Total area = 0.175 + 0.715 + 0.115 + 0.92 = 1.925 J
  - The best estimate is option **D**

Question 24 Hard [1 mark]

The graph below shows how force changes the extension of a spring.



The length of the spring with no force applied was 1 cm.

What is the **increase** in the strain energy stored in the spring when its **length** is increased from 2.0 cm to 3.0 cm ?

A 0.050 J

B 0.040 J

C 0.030 J

D 0.020 J

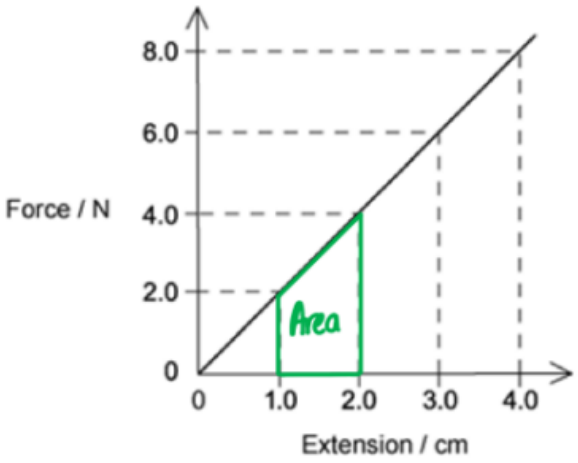
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The strain energy is the energy stored in an elastic body under loading
- The spring is elastic and obeying Hooke's law as the graph is proportional and goes through the origin
  - The spring has an original length of 1.0 cm
  - The strain energy is equal to the area under the force-extension graph.
- The question asks for the increase in length, not extension
- Original length = 1.0 cm
  - When length = 2.0 cm, extension $= 2.0 - 1.0 = 1.0$ cm extension

- When length = 3.0 cm, extension = $3.0 - 1.0 = 2.0$ cm

- The strain energy can be calculated from the area under the graph between 1.0 cm and 2.0 cm, 1.0 cm = 0.01 m

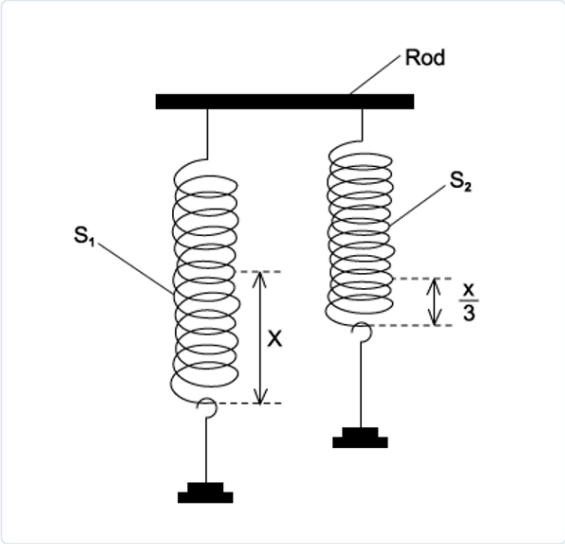


- Area of a trapezium $\frac{1}{2} \times \sum parallel \quad sides \times height$
- Therefore, the strain energy stored is $= \frac{1}{2} \times 2.0 + 4.0 \times 0.01 = 0.030$ J

Question 25 Hard [1 mark]

Two springs are attached to a rod, S_1 and S_2. A load is applied to the bottom of S_1, and it extends by *x* with an elastic potential energy in the spring of *E*.

The same load is applied to the bottom of S_2. The extension of S_2 is $\frac{x}{3}$, as shown in the diagram



Which equation shows the elastic potential energy in S_2?

A $9E$

B $3E$

C $\frac{E}{3}$

D $\frac{E}{9}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Elastic potential energy is the energy stored as a result of deformation of an elastic object
  - Elastic potential energy $= \frac{1}{2}Fx$
- The spring constant *k* of each of the springs is different as well as the extension *x* due to the load applied
- For spring S_1:
  - $E = \frac{1}{2}Fx$
  - $F = \frac{2E}{x}$

- For spring S_2:
  - $E_2 = \frac{1}{2}F\frac{x}{3} = \frac{1}{6}Fx$
- Since the load *F* applied to both springs is the same
  - $F = \frac{6E_2}{x}$
  - $F = \frac{2E}{x} = \frac{6E_2}{x}$
  - $E_2 = \frac{2E}{6} = \frac{E}{3}$

Therefore, if spring S_1 extends by *x*, then the energy stored in S_2 will be $\frac{E}{3}$